AQA 3.1.9 Rate equations

AQA A-level Chemistry · Revision questions · Sheet 6 of 6

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- Doubling a concentration: rate × 1 (zero order), × 2 (first order), × 4 (second order)
- $\ln k = -E_a/RT + \ln A$, so the gradient of ln k against 1/T = $-E_a/R$
- Use E_a in J mol^{-1} and T in K with R = 8.31 J K^{-1} mol^{-1}; convert to kJ mol^{-1} at the end
- Clock reactions: the same amount of product forms each time, so initial rate $\propto 1/t$
- rate = $k[A]^m[B]^n$ — orders come from experimental data, not from the balanced equation

Section A: Short-answer questions

01 The rate equation for a reaction is rate = $k[X]^2$. State the factor by which the rate increases when the concentration of X is tripled.

[1 mark]

02 State the meaning of the term *activation energy*.

[1 mark]

03 The Arrhenius equation is $k = Ae^{-Ea/RT}$. For a reaction, A = 2.0 × 10^{10} s^{-1} and E_a = 50.0 kJ mol^{-1}. Calculate the value of k at 300 K. The gas constant R = 8.31 J K^{-1} mol^{-1}

[2 marks]

04 Hydrogen peroxide solution decomposes to form water and oxygen: $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$. The reaction is catalysed by manganese(IV) oxide.

Explain why the reaction is fastest at the start.

[2 marks]

05 The rate equation for a reaction of A is rate = $k[A]^2$. Suggest the reactant(s) in the rate-determining step.

[1 mark]

Section B: Longer questions

06 Iodine and propanone react together in an acid-catalysed reaction.

$CH_3COCH_3(aq) + I_2(aq) \rightarrow CH_3COCH_2I(aq) + HI(aq)$

A student completed a series of experiments to determine the order of reaction with respect to iodine.

Method

- Transfer 25 cm^3 of 1.0 mol dm^{-3} propanone solution into a conical flask.
- Add 10 cm^3 of 1.0 mol dm^{-3} HCl(aq)
- Add 25 cm^3 of 5.0 × 10^{-3} mol dm^{-3} I_2(aq) and start a timer.
- At intervals of 1 minute, remove a 1.0 cm^3 sample of the mixture and add each sample to a separate beaker containing an excess of $NaHCO_3$(aq)
- Titrate the contents of each beaker with a standard solution of sodium thiosulfate and record the volume of sodium thiosulfate used.

06.1 Suggest why the 1.0 cm^3 portions of the reaction mixture are added to an excess of $NaHCO_3$ solution.

[2 marks]

06.2 Suggest why the order of this reaction with respect to propanone can be ignored in this experiment.

[2 marks]

06.3 A graph of the volume of sodium thiosulfate used (which is proportional to $[I_2]$) against time is a straight line. Explain how this shows that the reaction is zero order with respect to iodine.

[2 marks]

07 A general equation for a reaction is shown.

A(aq) + B(aq) + C(aq) → D(aq) + E(aq)

In aqueous solution, A, B, C and D are all colourless but E is dark blue.

A reagent (X) is available that reacts rapidly with E. This means that, if a small amount of X is included in the initial reaction mixture, it will react with any E produced until all of the X has been used up.

Explain, giving brief experimental details, how you could use a series of experiments to determine the order of this reaction with respect to A. In each experiment you should obtain a measure of the initial rate of reaction.

[6 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	9	*Source: Practice question (written for this set)*	1
02	The minimum energy (that particles need) for a reaction to occur / for a successful collision	*Source: Practice question (written for this set)*	1
03	M1 $k = 2.0 \times 10^{10} \times e^{-50\,000 \div (8.31 \times 300)}$ M2 $k = 39(.0)\ s^{-1}$	*Source: Practice question (written for this set)* If E_a not converted to J mol^{-1}, M1 only	2
04	M1 highest concentration of H_2O_2 / more particles / molecules M2 more frequent successful collisions	*Source: 7405/3 Jun 2022 Q04.1* Ignore 'chance' / 'probability'	2
05	Two molecules / particles of A (e.g. A + A → …)	*Source: Practice question (written for this set)*	1
06.1	M1 the sodium hydrogencarbonate neutralises the acid (catalyst) M2 so stops the reaction	*Source: 7405/2 Nov 2021 Q10.1, 10.2, 10.4*	2
06.2	M1 the concentration / amount of propanone is much larger than (200 times) the concentration / amount of iodine M2 concentration of propanone is (almost) constant		2
06.3	M1 the graph is a straight line / has a constant gradient M2 so the rate of reaction does not change as the concentration of iodine changes / iodine is used up at a constant rate	Original asks students to plot the graph first; stem adapted	2
07	**Level 3 (5–6 marks)** All stages covered and explanation of each stage correct and virtually complete; coherent with progression through all three stages. **Level 2 (3–4 marks)** All stages covered but explanation may be incomplete or contain inaccuracies, OR two stages covered and generally correct and virtually complete. **Level 1 (1–2 marks)** Two stages covered but incomplete / inaccurate, OR one stage covered and generally correct and virtually complete. **Level 0** Insufficient correct chemistry.	*Source: 7405/3 Jun 2018 Q01.8* **Stage 1 Preparation** 1a measure (known volumes of) reagents; 1b measure (known amount of) X; 1c into separate containers, A, B or C added last (not X last) **Stage 2 Procedure** 2a start timer at mixing; 2b time for appearance of blue colour; 2c same concentration of B and C / same total volume / same amount of X; 2d same temperature / water bath; 2e repeat with different concentrations of A **Stage 3 Results** 3a 1/time is a measure of rate; 3b plot 1/time against concentration of A (or log plot); 3c interpret order from shape / gradient, e.g. 2 × [A] gives 2 × rate so 1st order Alternative: colorimeter with continuous monitoring, gradient at t = 0	6